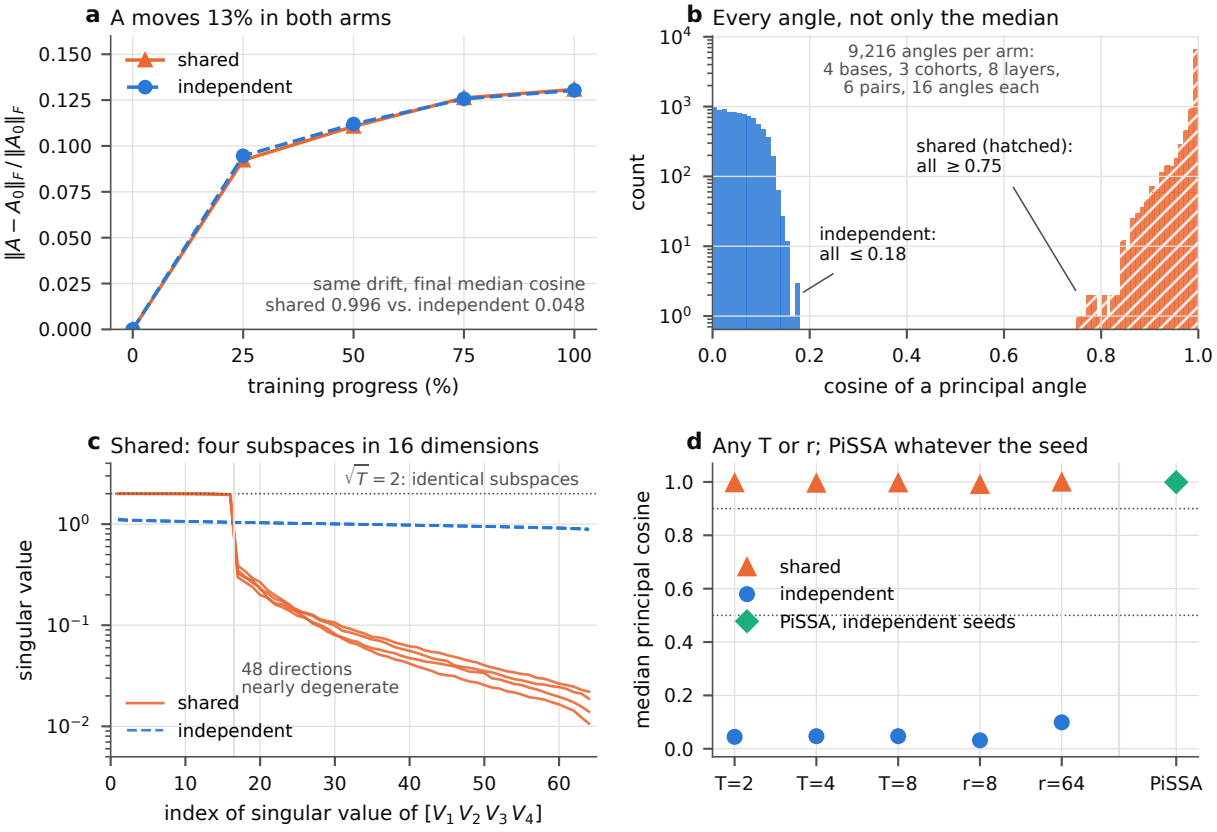
a A moves 13% in both arms
shared
independent
$\|A - A_0\|_F / \|A_0\|_F$
0.150
0.125
0.100
0.075
0.050
0.025
0.000
same drift, final median cosine
shared 0.996 vs. independent 0.048
0
25
50
75
100
training progress (%)
b Every angle, not only the median
9,216 angles per arm:
4 bases, 3 cohorts, 8 layers,
6 pairs, 16 angles each
shared (hatched):
all ≥ 0.75
independent:
all ≤ 0.18
count
10^4
10^3
10^2
10^1
10^0
0.0
0.2
0.4
0.6
0.8
1.0
cosine of a principal angle
c Shared: four subspaces in 16 dimensions
$\sqrt{T} = 2$: identical subspaces
singular value
10^0
10^{-1}
10^{-2}
48 directions
nearly degenerate
shared
independent
0
10
20
30
40
50
60
index of singular value of $[V_1\, V_2\, V_3\, V_4]$
d Any T or r; PiSSA whatever the seed
median principal cosine
1.0
0.8
0.6
0.4
0.2
0.0
shared
independent
PiSSA, independent seeds
T=2
T=4
T=8
r=8
r=64
PiSSA